
Custodial vs. Minimal: a 2×2 Comparison

One 2×2 grid per observable — **PAPER** vs. **OURS**, **no custodial** vs. **custodial**.

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1 The question, and the one-line answer

The question. Does custodial symmetry ($SU(2)_L \times SU(2)_R \times U(1)_X \times P_{LR}$) move the floors that our minimal-RS scan actually runs into? For each of three observables we put four panels on a strict 2×2 grid: the published figure (**PAPER**, left column) beside our fixed-code reproduction (**OURS**, right column), *without* custodial (top row) above *with* custodial (bottom row).

The one-line answer. Custodial is an *electroweak* cure, not a *flavor* cure. It pulls the electroweak floors down — the oblique T parameter collapses to ~ 0 and the $Z b\bar{b}$ shifts collapse onto the SM — but it leaves the ε_K wall flat at tens of TeV, because the binding kaon operator is the color-octet KK-gluon C_4 , an electroweak singlet the custodial group cannot touch. This is visible *both* in the literature and in ours: Blanke’s *full* custodial $\Delta F=2$ calculation finds the same ε_K wall as the minimal anarchic scans.

How to read each grid. Columns: **PAPER** (cropped published figure) | **OURS** (our code). Rows: **no custodial** (top) | **custodial** (bottom). For ε_K the two OURS panels are deliberately the *same* cloud (custodial changes nothing); for S, T, U and $Z \rightarrow b\bar{b}$ the bottom-row trajectory visibly relaxes toward the SM. The detailed provenance, attribution, and per-paper floor tables live in `custodial_comparison.tex`; this note is just the pictures.

Lane note. The ε_K comparison uses our **anarchic baseline (lane A)** — the literature strawman — so it lines up against Bauer/Blanke. Production as run (lane B, fit-aligned MFV) is a sharper wall at ≈ 7 TeV; the FPR/aligned ideal (lane C) is ≈ 2 TeV. Custodial moves *none* of these flavor lanes; only flavor alignment does.

2 (1) ε_K — the wall does *not* move

Custodial 2×2: ε_K (neutral-kaon CP violation)

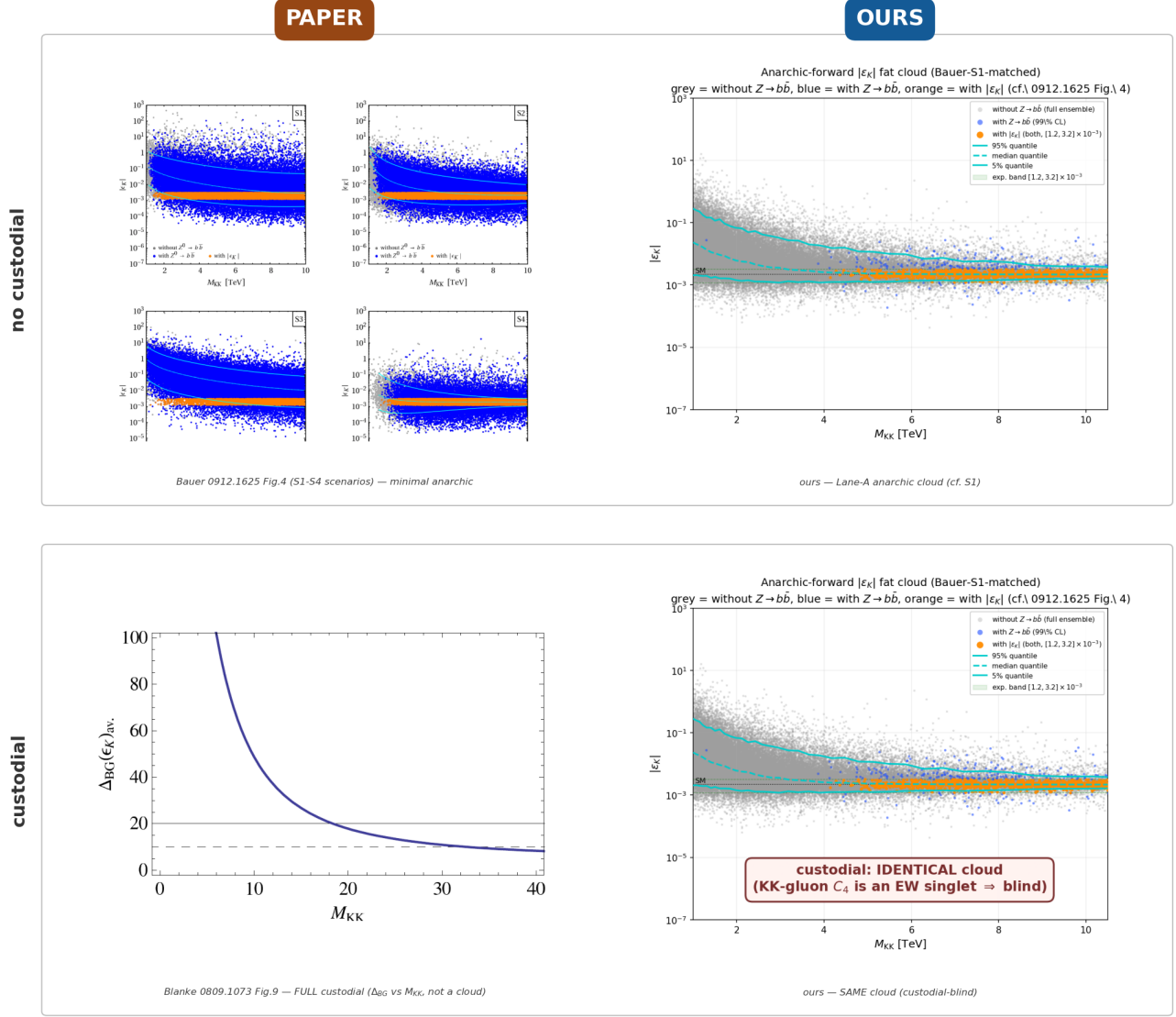


Figure 1: ε_K : custodial is blind to the kaon wall. *Top (no custodial)*: the minimal anarchic $|\varepsilon_K|$ cloud, Bauer 0912.1625 Fig. 4 (left) and our Lane-A reproduction (right) — both span ~ 6 decades with the same grey / blue / orange $Z \rightarrow b\bar{b}$ / $|\varepsilon_K|$ layering. *Bottom (custodial)*: Blanke 0809.1073 Fig. 9 (left) computes the ε_K fine-tuning **inside the full custodial model** and still crosses $\Delta_{BG} < 20$ only at $M_{KK} \approx 18$ TeV and $\Delta_{BG} < 10$ at ≈ 30 TeV; our custodial panel (right) is the *identical* cloud plus the annotation, because the KK-gluon C_4 operator is an EW singlet that custodial cannot touch. **The wall stays at tens of TeV in both the paper (Blanke, full custodial) and ours.** Floors: ours Lane-A ~ 9 –24 TeV; literature ~ 18 –33 TeV (Blanke 18/30, CFW 21/33, Bauer II ~ 10 , Archer 10–30); custodial moves none of them.

3 (2) S, T, U oblique — custodial pulls $T \rightarrow 0$

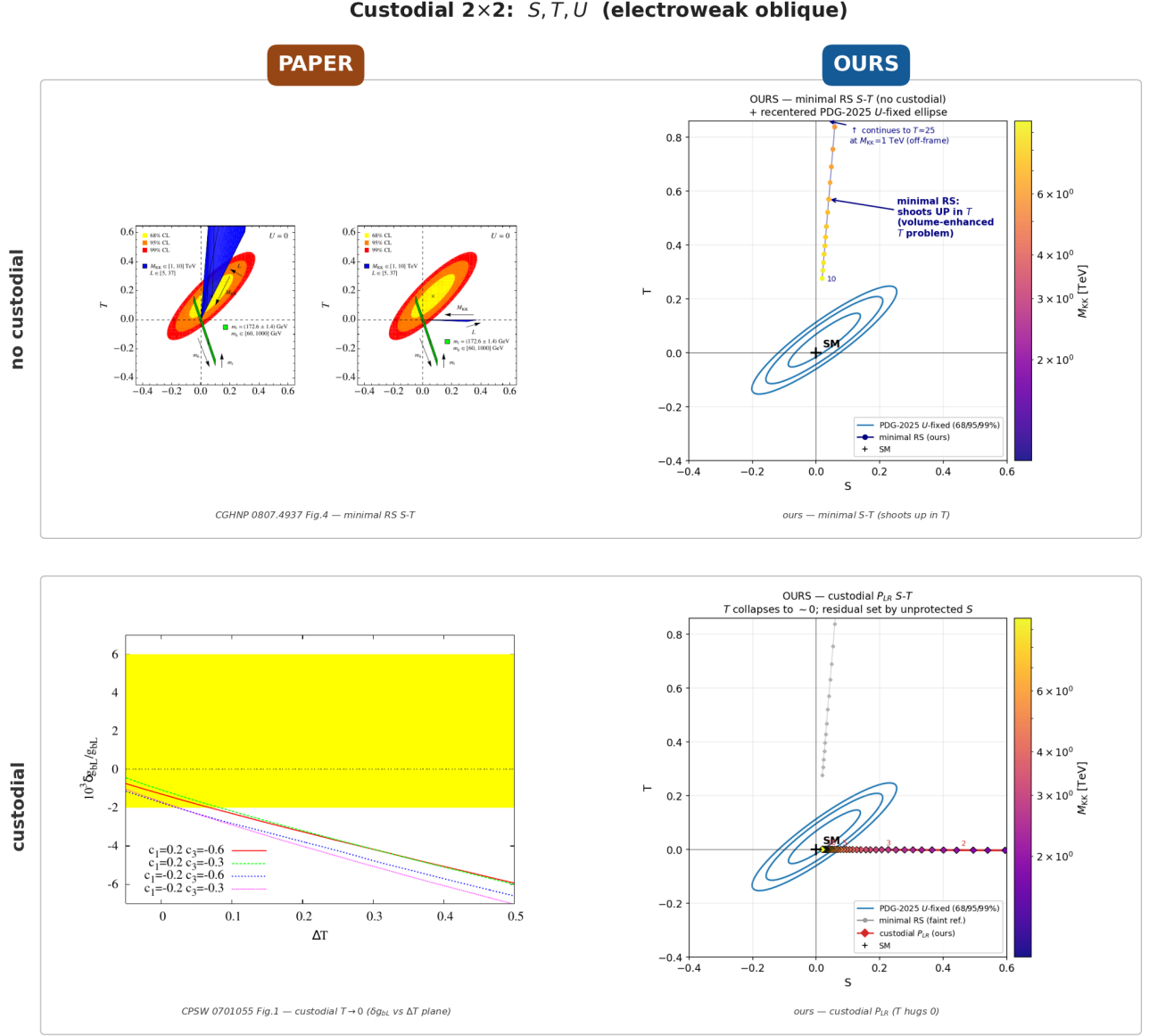


Figure 2: S, T, U : custodial protects T . *Top (no custodial):* the minimal RS S – T trajectory climbs steeply in T (the volume-enhanced “RS T problem”), in CGHNP 0807.4937 Fig. 4 (left) and in ours (right, the points shoot up off the recentered PDG-2025 ellipse). *Bottom (custodial):* CPSW 0701055 Fig. 1 (left) shows the custodial δg_{bL} – ΔT correlation with ΔT driven back toward 0; our custodial P_{LR} trajectory (right) hugs the S -axis at $T \approx 0$ (minimal T reaches ~ 25 at 1 TeV; custodial $|T| < 0.01$ everywhere), the ADMS mechanism ($SU(2)_R$ turns $\Delta T \propto +L$ into $-1/L$). **Custodial relieves the oblique floor in both paper and ours:** our minimal ~ 18 – 20 TeV existence wall $\rightarrow$ custodial ~ 2 – 3 TeV (literature, S -limited), with the residual set by the unprotected S .

4 (3) $Z \rightarrow b\bar{b}$ — custodial collapses onto the SM

Custodial 2×2: $Z \rightarrow b\bar{b}$ (b -quark Z couplings)

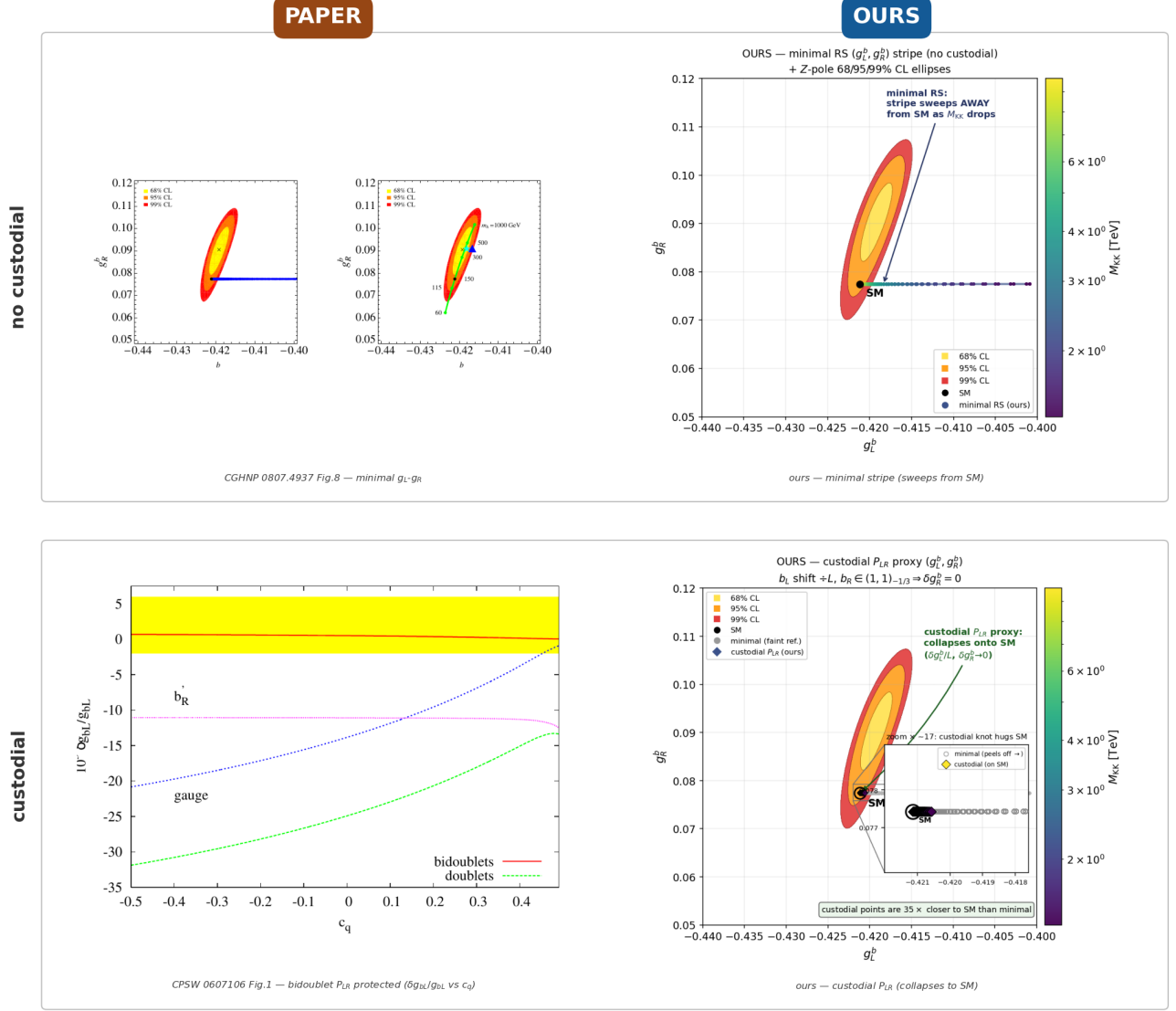


Figure 3: $Z \rightarrow b\bar{b}$: P_{LR} collapses the b couplings to the SM. *Top (no custodial):* the minimal (g_L^b, g_R^b) stripe sweeps away from the SM as M_{KK} drops, in CGHNP 0807.4937 Fig. 8 (left) and ours (right), overlaid on the Z -pole 68/95/99% CL ellipses. *Bottom (custodial):* CPSW 0607106 Fig. 1 (left) — the **bidoublet** embedding (red) sits inside the allowed band because P_{LR} with $b_L \in (2, 2)_{2/3}$ zeroes δg_L^b at tree level, while the unprotected “gauge” / “doublets” curves run far below; our custodial P_{LR} proxy (right) collapses onto the SM (δg_L^b suppressed by $1/L$, $\delta g_R^b \rightarrow 0$ for $b_R \in (1, 1)_{-1/3}$), landing $\sim 35\times$ closer to the SM than the minimal stripe. **Custodial protects $Zb_L b_L$ at tree level in both paper and ours** (top-partner loops partially re-introduce it; see `custodial_comparison.tex`). Floor: our minimal $Z \rightarrow b\bar{b}$ bites only to ~ 5 TeV (post-B1-fix; the old 25–30 TeV was a sign-bug).

5 Bottom line

- **Custodial is an EW cure, not a flavor cure.** It pulls $T \rightarrow 0$ and collapses the $Z b\bar{b}$ shifts onto the SM, but leaves the ε_K wall flat at tens of TeV — in the literature (Blanke’s full custodial $\Delta F=2$) and in ours.
- **The EW floors drop, the flavor wall does not.** S, T, U : $\sim 18\text{--}20$ TeV $\rightarrow \sim 2\text{--}3$ TeV (S -limited). $Z \rightarrow b\bar{b}$: tree-protected. ε_K : unchanged at tens of TeV; only flavor *alignment* (lane C) moves it to ~ 2 TeV.
- **These plots show the leading physics only.** The OURS custodial panels use a leading-order tree EW proxy (custodial T , P_{LR} $Z b_L \bar{b}_L$ protection); the subleading custodial phenomenology is the deferred full build, scoped in `custodial_comparison.tex`.

Two caveats on the OURS custodial panels (so they are not over-read).

1. *The $Z \rightarrow b\bar{b}$ custodial collapse ($\delta g_R^b \rightarrow 0$) is representation-specific.* It is exact only for $b_R \in (1, 1)_{-1/3}$, where $T_L^3 = T_R^3 = 0$ gives P_C protection (ACDP Table 1, row 2). Other b_R embeddings give a *nonzero* δg_R^b of either sign (e.g. the $(2, 4)$ -type used to fit A_{FB}^b). Our panel pins the protected embedding; it is a model choice, not a theorem.
2. *The “ ε_K identical” claim is exact for the dominant operator, not for every flavor observable.* The ε_K wall is genuinely custodial-blind — it is set by the color-octet KK-gluon C_4 , an EW singlet (and Blanke’s full custodial $\Delta F=2$ confirms it). But custodial *does* touch flavor at subleading order through the EW KK bosons (Blanke: Z_H, Z' can compete with the KK gluon in B -mixing) and custodian loops ($b \rightarrow s\gamma$). Our `deltaf2` path has no `ew_model` branch, so “custodial changes *nothing* in flavor” is a leading-order statement; what is rigorous is that the binding ε_K wall does not move.